CHAPTER 7

Alternating Current

Mean Value and Virtual Value of A.C.

1. A standard filament lamp consumes 100 W when connected to 200 V ac mains supply. The peak current through the bulb will be: *(2022 Re)*

a. 2 A b. 0.707 A c. 1A d. 1.414

2. The peak voltage of the **ac source** is equal to *(2022)*

a. $\frac{1}{\sqrt{2}}$ times the rms value of the source

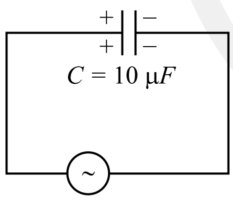
b. the value of voltage supplied to the circuit

c. the rms value of the ac source

d. $\sqrt{2}$ times the rms value of the ac source

A.C. Circuit Containing Capacitance Only

3. A10mF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly (p = 3.14): *(2024)*



210 *V*, 50 *HZ*

a. 1.20 A b. 0.35 A c. 0.58 A d. 0.93 A

4. Given below are two statements. *(2022 Re)*

Statement I: In an ac circuit, the current through a capacitor leads the voltage across it.

Statement II: In a.c. circuits containing pure capacitance only, the phase difference between the current and the voltage is π.

In the light of the above statements, choose the most appropriate answer from the options given below

a. Statement I is incorrect but Statement II is correct

b. Both Statement I and Statement II are correct

c. Both Statement I and Statement II are incorrect

d. Statement I is correct but Statement II is incorrect

5. A capacitor of capacitance 'C', is connected across an ac source of voltage V, given by $V = V_0 \sin \omega t$. The displacement current between the plates of the capacitor, would then be given by: *(2021)*

a. $I_d = \frac{V_0}{\omega C} \cos \omega t$ b. $I_d = \frac{V_0}{\omega C} \sin \omega t$

c. $I_d = V_0 \omega C \sin \omega t$ d. $I_d = V_0 \omega C \cos \omega t$

6. A 40 μF capacitor is connected to a 200V, 50Hz ac supply. The rms value of the current in the circuit is, nearly: *(2020)*

a. 2.05 A b. 2.5 A c. 25.1 A d. 1.7 A

7. A small signal voltage $V(t) = V_0 \sin\omega t$ is applied across an ideal capacitor C : *(2016 - I)*

a. Current I(t) lags voltage V(t) by 90°

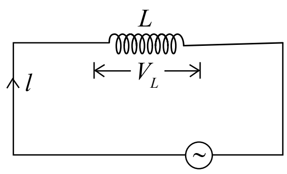
b. Over a full cycle the capacitor C does not consume any energy from the voltage source

c. Current I(t) is in phase with voltage V(t)

d. Current I(t) leads voltage V(t) by 180°

A.C. Circuit Containing R and L Only

8. In the circuit shown below, the inductance *L* is connected to an ac source. The current flowing in the circuit is $I = I_0 \sin wt$. The voltage drop (V_L) across *L* is *(2024 Re)*



ac source

a. $wL\, l_0 \sin wt$ b. $\frac{I_0}{\omega L} \sin\omega t$

c. $\frac{I_0}{\omega L} \cos\omega t$ d. $wL\, l_0 \cos wt$

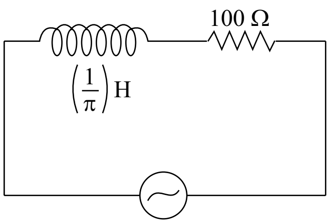
9. The amplitude of the charge oscillating in a circuit decreases exponentially as $Q = Q_0 e^{-Rt/2L}$, where Q_0 is the charge at t = 0 *s*. The time at which charge amplitude decreases to 0.50 Q_0 is nearly:

[Given that *R* = 1.5 W, *L* = 12 mH, *ln*(2) = 0.693] *(2024 Re)*

a. 19.01 ms b. 11.09 ms

c. 19.01 s d. 11.09 s

10. An ac source is connected in the given circuit. The value of ϕ will be: *(2023-Manipur)*



$V = 220 \sin(100\pi t + \phi)$ volt

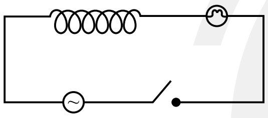
a. 60° b. 90° c. 30° d. 45°

11. An inductor of inductance 2 mH is connected to a 220 V, 50 Hz a.c. source. Let the inductive reactance in the circuit is X_1. If a 220 V dc source replaces the ac source in the circuit, then the inductive reactance in the circuit is X_2.

X_1 and X_2 respectively are *(2022 Re)*

a. 0.628 Ω, infinity b. 6.28 Ω, zero
c. 6.28 Ω, infinity d. 0.628 Ω, zero

12. A light bulb and an inductor coil are connected to an ac source through a key as shown in the figure below. The key is closed and **after sometime** an iron rod is inserted into the interior of the inductor. The glow of the light bulb *(2020-Covid)*



a. Remains unchanged b. Will fluctuate
c. Increases d. Decreases

13. A resistance R draws power P when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes Z, the power drawn will be: *(2015)*

a. $P\sqrt{\frac{R}{Z}}$ b. $P\left(\frac{R}{Z}\right)$ c. P d. $P\left(\frac{R}{Z}\right)^2$

14. A coil of self-inductance L is connected in series with a bulb B and an AC source. Brightness of the bulb decreases when: *(2013)*

a. An iron rod is inserted in the coil
b. Frequency of the AC source is decreased
c. Number of turns in the coil is reduced
d. A capacitance of reactance $X_C = X_L$ is included in the same circuit

A.C. Circuit Containing R and C Only

15. A series RC circuit is connected to an alternating voltage source. Consider two situations

A. When capacitor is air filled.
B. When capacitor is mica filled.

Current through resistor is i and voltage across capacitor is V then: *(2015 Re)*

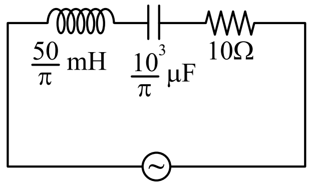
a. $V_a = V_b$ b. $V_a < V_b$ c. $V_a > V_b$ d. $I_a > I_b$

A.C. Circuit Containing R, L and C

16. In a series LCR circuit, the inductance L is 10 mH, capacitance C is 1μF and resistance R is 100Ω. The frequency at which resonance *(2023)*

a. 1.59 kHz b. 15.9 rad/s c. 15.9 kHz d. 1.59 rad/s

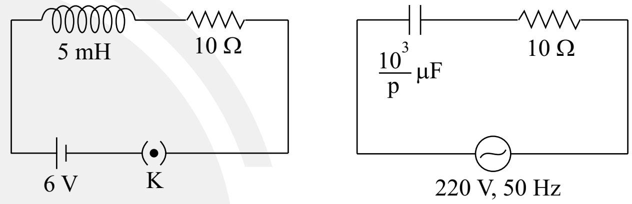
17. The net impedance of circuit (as shown in figure) will be: *(2023)*



220V, 50Hz

a. 25 W b. $10\sqrt{2}\ \Omega$ c. 15 W d. $5\sqrt{5}\ \Omega$

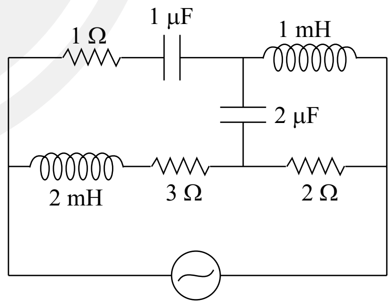
18. If Z_1 and Z_2 are the impedances of the given circuits (*a*) and (*b*) as shown in figures, then choose the **correct** option *(2023-Manipur)*



(a) (b)

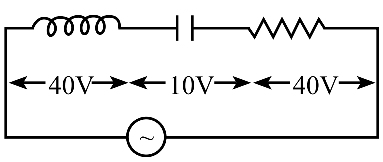
a. $Z_1 < Z_2$ b. $Z_1 + Z_2 = 20$ w
c. $Z_1 = Z_2$ d. $Z_1 > Z_2$

19. For every high frequencies, the effective impedance of the circuit (shown in the figure) will be: *(2023-Manipur)*



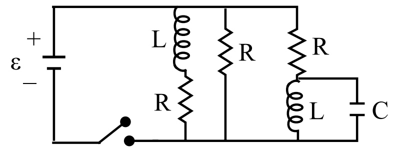
a. 4 W b. 6 W c. 1 W d. 3 W

20. An inductor of inductance L, a capacitor of capacitance C and a resistor of resistance 'R' are connected in series to an ac source of potential difference 'V' volts as shown in figure. Potential difference across L, C and R is 40V, 10V and 40V, respectively. The **amplitude** of current flowing through LCR series circuit is $10\sqrt{2}$ A. The impedance of the circuit is: *(2021)*



a. $5/\sqrt{2}\ \Omega$ b. 4 Ω c. 5 Ω d. $4\sqrt{2}\ \Omega$

21. Figure shows a circuit that contains three identical resistors with resistance R = 9.0 Ω each, two identical inductors with inductance L = 2.0 mH each, and an ideal battery with emf ε = 18 V. The current 'i' through the battery **just after** the switch closed is: *(2017-Delhi)*



a. 0.2 A b. 2 A c. 0 ampere d. 2 mA

Electric Resonance

22. A series LCR circuit with inductance 10 H, capacitance 10 μF, resistance 50 Ω is corrected to an ac source of voltage, **V = 200 sin(100t)** volt. If resonant frequency of the LCR circuit is ν_0 and the frequency of the ac source is ν, then: *(2022)*

a. $\nu = 100\text{Hz}; \nu_0 = \frac{100}{\pi}\text{Hz}$ b. $\nu_0 = \nu = 50$ Hz

c. $\nu_0 = \nu = \frac{50}{\pi}\text{Hz}$ d. $\nu_0 = \frac{50}{\pi}\text{Hz}, \nu = 50\,\text{Hz}$

23. A series LCR circuit is connected to an ac voltage source. When L is removed from the circuit, the phase difference between current and voltage is $\frac{\pi}{3}$. If instead C is removed from the circuit, the phase difference is again $\frac{\pi}{3}$ between current and voltage. The power factor of the circuit is: *(2020)*

a. 0.5 b. 1.0 c. –1.0 d. Zero

24. Which of the following combinations should be selected for **better tuning** of an L-C-R circuit used for communication? *(2016 - II)*

a. R = 15 Ω, L = 3.5 H, C = 30 μF
b. R = 25 Ω, L = 1.5 H, C = 45 μF
c. R = 20 Ω, L = 1.5 H, C = 35 μF
d. R = 25 Ω, L = 2.5 H, C = 45 μF

Energy Stored in Condenser, Inductor and LC Oscillations

25. The magnetic energy stored in an inductor of inductance 4 μH carrying a current of 2 A is: *(2023)*

a. 8 μJ b. 4 μJ c. 4 mJ d. 8 mJ

Average Power and Power Associated with A.C. Circuits and Power Factor

26. The maximum power is dissipated for an ac in a/an: *(2023-Manipur)*

a. resistive circuit b. LC circuit
c. inductive circuit d. capacitive circuit

Transformer

27. A step up transformer is connected to an ac mains supply of 220 V to operate at 11000 *V*, 88 watt. The current in the secondary circuit, ignoring the power loss in the transformer, is *(2024 Re)*

a. 8 mA b. 4 mA
c. 0.4 A d. 4 A

28. In an ideal transformer, the turns ratio is $\frac{N_P}{N_s} = \frac{1}{2}$ The ratio $V_s : V_p$ is equal to (the symbols carry their usual meaning): *(2024)*

a. 1:1 b. 1:4
c. 1:2 d. 2:1

29. A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to ac mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding? *(2023)*

a. 0.37 A b. 0.27 A
c. 2.7 A d. 3.7 A

30. A step down transformer connected to an ac mains supply of 220 V is made to operate at 11 V, 44 W lamp. **Ignoring power losses** in the transformer, what is the current in the primary circuit? *(2021)*

a. 0.4 A b. 2 A
c. 4 A d. 0.2 A

31. A transformer having efficiency of 90% is working on 200 V and 3 kW power supply. If the current in the secondary coil is 6 A, the voltage across the secondary coil and the current in the primary coil respectively are: *(2014)*

a. 300 V, 15 A b. 450 V, 15 A
c. 450 V, 13.5 A d. 600 V, 15 A

Answer Key

1. (b)	**2.** (d)	**3.** (d)	**4.** (d)	**5.** (d)	**6.** (b)	**7.** (b)	**8.** (d)	**9.** (b)	**10.** (d)
11. (d)	**12.** (d)	**13.** (d)	**14.** (a)	**15.** (c)	**16.** (a)	**17.** (d)	**18.** (a)	**19.** (d)	**20.** (c)
21. (None)	**22.** (c)	**23.** (b)	**24.** (a)	**25.** (a)	**26.** (a)	**27.** (a)	**28.** (d)	**29.** (b)	**30.** (d)
31. (b)									